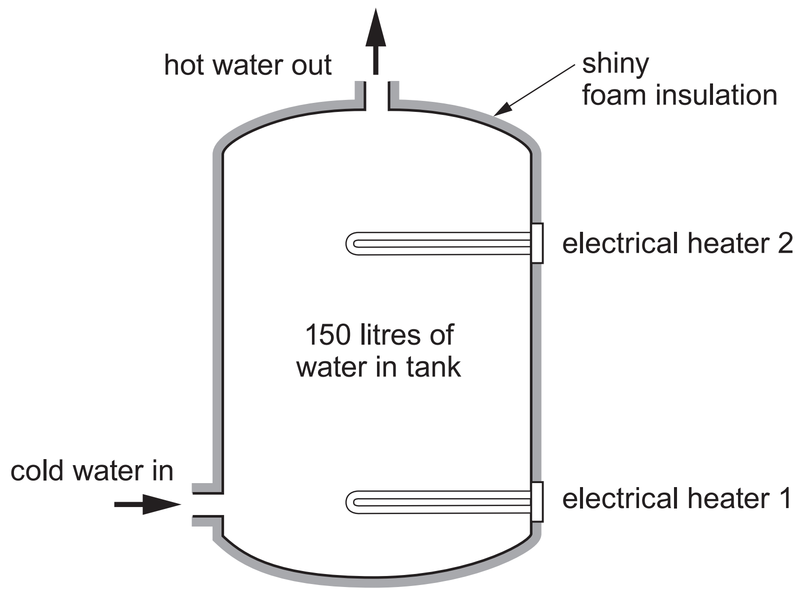
Examiner only

6. Paul lives in an apartment in central London. His electricity bill is £80 a month. Paul wants to replace his old electric boiler with a new one. A local plumbing company has suggested using a new boiler, the Redstar boiler, which has two electrical heaters. Paul will be able to use either heater to provide hot water after the boiler is installed.

Information about the Redstar boiler



Savings

Cost of boiler	£2500.00
Typical savings	£15.00 per month

Technical information on the two heaters in the Redstar boiler

Information	heater 1	heater 2
maximum volume of water that is heated by heater (litres)	150	50
time to heat this volume of water (hours)	3	0.5
total power supplied (kW)	2	4
power used to heat the water (kW)	1.6	3.6
efficiency (%)	80	
cost to heat the water (p)		44

(a) Use the information given about the boiler to answer the following questions.

(i) Calculate the expected payback time for the boiler in years. [3]

payback time = ... years

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(ii) Use the equation:

$$\%\ \text{efficiency} = \frac{\text{power usefully transferred}}{\text{total power supplied}} \times 100$$

to calculate the efficiency of **heater 2**. [2]

% efficiency = ...

(iii) Use the equations:

units used (kWh) = power (kW) × time (h)

total cost = cost of one unit × units used

to calculate the cost of the electricity to heat 150 litres of water using **heater 1**.

Cost per unit of electricity = 22 p [3]

cost = ...

(iv) Paul has estimated that he uses 40 litres of hot water per day.
He thinks that **heater 1** would be the best one to use every day.

Explain whether you agree with Paul. [2]

....

....

....

Examiner only

(b) The water in London is known to be hard.

Explain how hard water could have affected Paul's old boiler. [2]

....

....

....

(c) Heat energy can be transferred by conduction, convection and radiation.

(i) Explain how the shiny foam insulation used around the boiler helps to reduce heat loss. [2]

....

....

....

....

(ii) Explain how having the foam insulation layer around the boiler benefits the environment. [2]

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....

(iii) Paul thinks that **heater 2** will not heat all the water in the boiler.

Use the diagram of the boiler to explain whether you agree. [3]

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END OF PAPER

